

10th International Conference of Information and Communication Technology (ICICT-2020)

Parking diversion method based on Internet of Things and multi-word combination

Jing Jing He, Hao Tian Li, Ling Xue Wang, Yuan Wang*

*College of Artificial Intelligence Tianjin University Science and Technology,
Thirteenth Avenue Binhai New District Tianjin, Tianjin 300450, China*

Abstract

With the continuous improvement of living standards, motor vehicles have become an indispensable transportation in our daily life. The number of cars in each city continues to rise, causing variable degrees of parking difficulties in different regions. It has become a common phenomenon that no parking spot near popular places. With this, there are long queues of vehicles in front of the parking lot, which delays the time of car owners and causes traffic jams. This article introduces a new parking diversion method, based on the support vector machine (SVM) and simulated annealing fruit fly optimization algorithm forecasting models to predict short-term traffic flow near the parking lot. Through predictions, recommending parking lots with low traffic flow for car owners. Meanwhile, the method based on the Internet of Thing to lock pots, let vehicles enter according to the queue number and so on. Through this way, achieving online queuing and advance reservation. This method can solve the queuing phenomenon in front of the parking lots and give car owners the most suitable travel time, therefore, park will be more orderly and diversion will be realized.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: new parking diversion method; SVM; forecasting model; simulated annealing fruit fly optimization algorithm; The Internet of Things; online queuing; advance reservation.

* Corresponding author. Tel.: 022-60600983.

E-mail address: wangyuan23@tust.edu.cn

1. Introduction

Nowadays, the number of automobiles in our country has increased substantially, parking spaces in short supply now. The scenes of long queues of vehicles waiting in front of parking lots near the crowded places are often seen, which causes car owners to waste their time and road congestion. The paper proposes solutions to ease this phenomenon by selecting parking lot and booking parking spots in advance. Before this paper, the existing research has been quite perfect for traffic flow forecasting. There are a lot of different algorithms and models for forecasting the short-term traffic flow such as the method based on the SVM¹, use deep learning², use wavelet analysis³. Besides, there are some researches on vehicle reservations. However, the parking diversion method proposed for the blocking near the depot is unusual, and it lacks a way to ensure that each vehicle can get parking lot at the fastest speed while ensuring the diversion of vehicles. This method predicts the traffic flow outside the parking lot and provides car owners with better parking options for diversion. Meanwhile, the method provides different options for reserved empty parking spaces and cloud queuing according to whether the parking spaces are full, so that car owners own parking spots before they get. According to the Internet of Things, drivers can immediately enter the parking lot when they arrive at it. For traffic flow prediction, this method uses SVM and simulated annealing fruit fly optimization algorithm to analyze short-term traffic flow at the intersection near the depots. Before departure, you can reserve an empty parking space with vacant parking spaces or queue online in the full depot. After you choosing the depot, the server is sent to the sensing device so that the parked without additional queues. Now, the number of vehicles in each city continues to rise steadily, but the supply of parking spaces has not kept up. Traffic congestions in front of the parking lots are everywhere. This method can alleviate jams by selecting parking lots with less traffic flow before arrival and selecting parking spaces in advance which is beneficial for the current situation and create a good reference for researchers to study these problems.

2. Related Work

Now, the researchers did not delve into solving the traffic jam caused by parking, they are only interested in the reasons. Driving characteristics of exit vehicles during peak hours in large parking lots are investigated and selected two indicators: the exit time distance for vehicles inline and the exit service time. An estimate of the M/G/I queuing model is derived from the measurement characteristics of these two indicators⁴. There is also some analysis on the service efficiency of entrance lanes and charging standards⁵. Nowadays, numerous researchers have discussed the two things in predicting traffic flow and booking parking spaces. For predicting traffic flow, forecasting methods have two types: One is using traditional mathematical and physical methods such as mathematical statistics and calculus to build a model to predict. This includes the TimeSeries Model, Parametric Regressive Model, Exponential Smoothing Model, etc. The other is study of modern technologies and methods (such as simulation technology, neural network, fuzzy control). This kind of method pursues the fitting effect of the real traffic flow phenomenon. This category includes Nonparametric Regressive Model, KARIMA algorithm, wavelet theory-based methods, multi-dimensional fractal-based methods, Spectral Basis Analysis, state-space reconstruction models, and multiple composite prediction models related to Neural Network. However, there are currently four widely-used models. Their names, merits, and demerits are as shown in the table below⁶:

Table 1. Four widely used models

Model	Advantage	Disadvantage
History Average Model	Easy to use, high speed	Unable to cope with emergencies
Time-Series Model- ARIMA	Mature technology, higher accuracy	Data loss, complex initialization
Neural Network Model	Suit for complex nonlinear conditions	Complex training process
Nonparametric Regressive Model	No need to assume basic relationships	Complexity of finding close neighbors

These models have their own merits and demerits. And the data is not optimized and filtered. Most of the solutions to the parking space reservation problem rely on the Internet of Things and the Internet of Vehicles, and

combine functions such as parking space indexing and vehicle positioning in the parking lot. However. The cloud queuing in disguised form of appointment is not considered.

3. Method

3.1. Support vector machine (SVM) and simulated annealing fruit fly optimization algorithm forecasting models

A supervised learning model and related learning algorithms, which are called SVM, regression analysis and analyzing data in classification. It can handle regression problem (time series analysis) and pattern recognition (time series analysis) and pattern recognition (classification problems, discriminant analysis) and many other problems, and can be promoted, forecasted, and comprehensively evaluated in other fields and disciplines⁷. When using SVM to make predictions, autoregressive methods are used to predict future values using previous data in the same time series. It is called an SVM regression machine, when SVM is used for regression estimation. The SVM regression machine can be expressed as⁸:

$$\begin{aligned} \min \frac{1}{2} \|k\|^2 + C \sum_{i=1}^l (a_i + a_i^*). \\ f(x_i, k) - y_i \leq X + a_i, i = 1, \dots, l; \\ y_i - f(x_i, k) \leq X + a_i^*, i = 1, \dots, l; \\ a_i \geq 0, a_i^* \geq 0, i = 1, \dots, l; \end{aligned} \quad (1)$$

Since commuters are the largest group of car owners on the road, their commuting time and rest time are basically the same every week. The traffic flow on weekends is greater than that on working days, in the meantime. Therefore, the weekly cycle is adopted to calculate the data flow on Monday, Tuesday...Sunday, and establish a prediction model:

$$X(t) = \sum_{i=1}^l (T_i^* - T_i) K(x_i, x) + b. \quad (2)$$

If there are $n+2$ data in a sequence, use $X_k(1), X_k(2) \dots X_k(n-1)$ as the input of model training, $X_k(2), X_k(3) \dots X_k(n)$ as Corresponding output, through training to determine the parameter model⁹ which is also based on simulated annealing fruit fly optimization algorithm. The figure below shows the fruit fly optimization's principle:

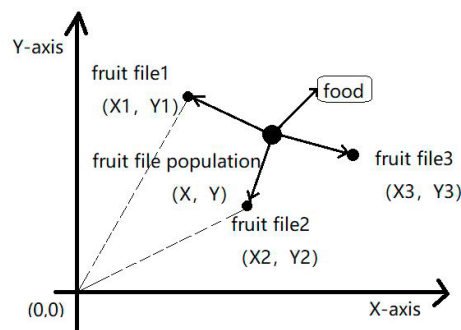


Fig. 1. The principle of fruit fly optimization

The principle of fruit fly optimization: the position (x, y) of the initial fruit fly population at random. The fruit flies use this position as the starting point to fly randomly to find food. When the individual fruit fly finds the best position for food and other fruit flies will also gather to this position (x2, y2); and then the next optimization starting point becomes a new location until the food is found. The number of iterations of the algorithm increases, fruit flies will increasingly approach the position of food. However, the step size setting has drawbacks that setting a small step size in the early stage causes low efficiency, and setting a larger value in the later stage misses the optimal solution. The efficiency and accuracy of the optimization step formula to a improved certain extent. It improves the step size as follows:

$$L' = n * \left(1 - (g - 1) / (\max gen)^{\lambda}\right) * \text{sgn}(0.5 - r) \quad (3)$$

From the formula in which step size is improved, the step size is in inverse proportion to the current iteration number. As the step size increase, the self-sufficiency decreases small. The improved algorithm steps are:

1) Initialize the position of fruit fly population (X_axis, Y_axis), and set the population size and the number of iterations maxgon;

2) Initialize the fruit fly population's direction and distance to find food;

$$\begin{cases} X_i = X_axis + L \\ Y_i = Y_axis + L \end{cases} \quad (4)$$

3) Calculate the taste concentration determination value which is substituted into the taste concentration judgment function, and find the individual taste concentration (smell) of the fruit fly;

$$s_i = 1 / \left[(X_i)^2 + (Y_i)^2 \right]^{\frac{1}{2}} \quad (5)$$

4) Find the fruit flies with the smallest taste concentration in the fruit fly population, record it as S(i). Keep the model position coordinates (Xj, Yj);

$$\text{Smell} = \text{Function}(s_i) \quad (6)$$

5) Perform a certain number of disturbances to the current position to obtain a new model again, and mark the position coordinates as (Xj', Yj');

$$\begin{cases} X_i' = X_i + L' \\ Y_i' = Y_i + L' \end{cases} \quad (7)$$

6) Calculate the difference in taste concentration between the previous and current models. It is accepted that if the concentration difference is under or equal to zero. If not, calculate the probability $p = \exp(-\Delta s / g)$, and if $p > r$ (r is a random number in $[0,1]$), the new model is accepted. At the same time, the new position is the beginning of the next optimization, otherwise, the prototype position coordinates are used for the next optimization¹⁰.

3.2. Reserve parking spaces and “cloud queue”

After predicting the traffic flow in the parking lot to provide a suitable parking lot, drivers can issue reservations for empty parking spaces through the terminal or send cloud queuing requests for full parking lots. For reservations,

car owners check the free parking spaces through the terminal, make reservations, and mark the free parking spaces. The system will restrict other vehicles from parking if the parking spaces are marked. When the car owners arrive at their reserved parking spaces, they can operate through the mobile terminal to lift the restriction. For cloud queuing, car owners select the full parking lot through the terminal and queue up which means make an appointment to enter. The overall system uses ZigBee for data collection, with Arduino control hardware. Online processing of data storage and data development, and with mobile devices such as mobile phones. The specific system framework is as follows:

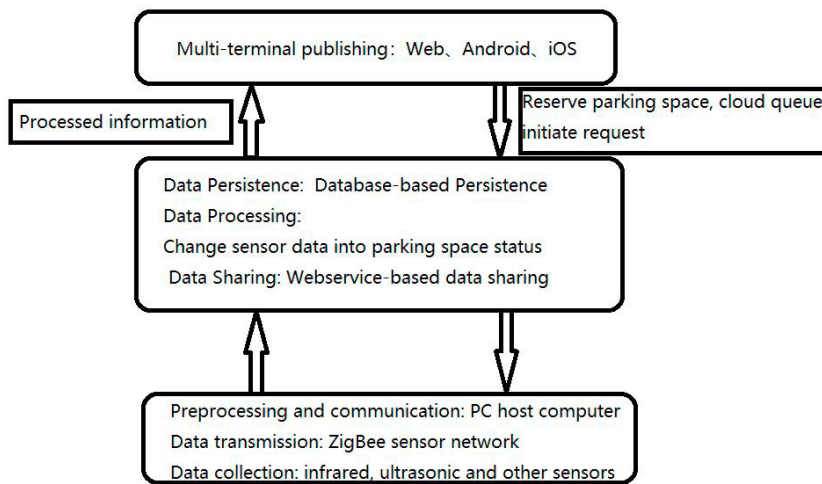


Fig. 2. The specifics framework

The ultrasonic sensor detects the distance to determine whether there is a vehicle getting into the park. The Arduino controller operates the data collected by the sensor and transmits it to the PC via ZigBee, finally, the data is passed to the server. Ultrasonic, Arduino controller, and ZigBee all use serial communication protocol to exchange data¹¹.

4. Conclusion

To solve the difficult parking problem caused by the increase in car ownership and the traffic congestion caused by long queues of vehicles near the depot, forming this model. A model constituted of SVM and fruit fly optimization algorithm based on simulated annealing is used to predict the traffic flow in front of each parking lot, giving car owners more information to choose a more suitable parking lot and more suitable travel time. At the same time, before departure, car owners can choose to reserve a parking space in advance, or queue in advance in a full parking lot. Send requests through the mobile terminal, use ZigBee for data collection, and collaborate in the Arduino controller to control the hardware, and process data storage and data development online. Ultrasonic sensors detect the distance to determine if a vehicle is entering.

The data is collected by the sensor and operated by the Arduino controller. Then it transmits to the PC through the ZigBee network, and ends with data transmission to the server. This method can efficaciously divert the flow and come up with a corresponding solution which is proposed for the queue near the parking lot. Providing car owners with a more suitable choice and making cars directly get into parking lots which eliminating the queue time and alleviating the phenomenon of long queues. Thinking about this phenomenon and proposing methods can also lay a firm foundation for subsequent research.

Acknowledgement

This paper is supported by College Student Innovation And Entrepreneurship Training Program: Parking reservation management and diversion APP. Program number: 202010057196.

References

1. Yang Zhaoshen: Short-term traffic flow prediction method based on support vector machine method, *Journal of Jilin University (Engineering Edition)*, 2006
2. Luo Xianglong: Short-term traffic flow prediction based on deep learning, *Computer Application Research*, 2016
3. Dou Huili: Traffic flow prediction method based on wavelet analysis and ARIMA model, *Journal of Tongji University (Nature Science Edition)*, 2009
4. Liu Junjie, Huo Guang, Liu Yubo: The queuing delay at parking lot exit based on M/G/I model, *Journal of transport science and engineering*, 2015
5. Li Rong: Research on the setting of entrance and exit lanes of extra-large parking lots (garages) based on queuing theory, *China Urban Transport Planning Annual Conference and the 28th Academic Seminar*, 2015
6. Liu Jing, Guan Wei: A Summary of Traffic Flow Forecasting Methods, *Journal of highway and transportation research and development* 2004
7. Ding Shifei, Qi Bingjuan, Tan Hongyan: An Overview on Theory and Algorithm of Support Vector Machines, *Journal of University of Electronic Science and Technology of China*, 2011
8. Deng Naiyang, Tian Yingjie: *New Method in Data Mining Support Vector Machine*[M] Beijing: Science Press, 2004
9. Geng Rui, Cui Deguang, Xu Bing: Support vector machine-based combinational model for air traffic forecasts, *Journal of Tsinghua University (Natural Science Edition)*, 2008
10. Zhang Bing, Zhang Damin, AMinghan: Drosophila optimization algorithm based on simulated annealing, *Journal of computer applications*, 2016
11. Wang Zhongsheng, Han Kang: Research of intelligent parking system based on ZigBee, *Microcomputer and Application*, 2016